FOUNDATIONS

Classical ciphers

From secret writing to systematic cryptanalysis

An introduction to cryptography

Five hand ciphers. Five ways to see what secrecy leaves behind.

LECTURE 01

Master’s level • Computer science & engineering

Three hours • Worked examples • Two breaks

FOUNDATIONS

The question for today

You intercept a message. You know the encryption method, but not the key.

INTERCEPTED

`PHHWPHDWQRRQ`

What information would let you recover the message?

Think about the attacker’s information before choosing an attack.

What you will be able to do

- Encrypt and decrypt with the five ciphers in this lecture.
- Use frequency, repetition, and position as evidence in an attack.
- Explain why a large key space alone does not establish security.
- Separate a historical teaching cipher from a tool suitable for real data.

Our three-hour route

Elapsed time	Focus
00–50 min	Foundations, shift, substitution
50–60 min	Break
60–110 min	Frequency analysis and Vigenère
110–120 min	Break
120–170 min	Playfair and transposition
170–180 min	Synthesis and exit questions

Exercises and discussion are included in these timings.

Five ciphers, three design ideas

Design idea	Ciphers in this lecture	What changes?
One symbol at a time	Shift; monoalphabetic substitution	Letter identity
Change by position / by pair	Vigenère; Playfair	Mapping with context
Rearrange a block	Transposition (permutation)	Letter positions

For each cipher: define a key, work an example, and identify what remains visible.

Why study ciphers we should not deploy?

- Small examples make the key, algorithm, and attacker visible.
- Failures expose recurring problems: predictable data and repeated structure.
- Manual attacks introduce the logic behind automated cryptanalysis.

Historical relevance does not imply present-day security.

A short historical thread

Period	Milestone
Antiquity	Caesar uses a fixed alphabet shift in correspondence.
9th century	Al-Kindi describes frequency-based cryptanalysis.
1553	Bellaso describes a repeating-key polyalphabetic scheme.
1854	Wheatstone invents the cipher promoted by Playfair.
19th century	Kasiski publishes a method attacking periodic substitution.

Historical overview: original deck; Kahn, The Codebreakers; Kasiski (1863).

The basic communication model



Plaintext is the original message; ciphertext is the encrypted message.

A cipher is the algorithm. A key selects one transformation from the family.

Correctness: decrypting an encrypted message with the right key recovers it.

Algorithm ≠ key

PUBLIC METHOD

```
shift each letter by K
```

Assume the attacker knows the algorithm.

SECRET KEY

```
K = 3
```

The key is the information the sender and receiver must protect.

Kerckhoffs’s principle: security should not depend on hiding the method.

Kerckhoffs, La Cryptographie Militaire (1883), principle 2.

Shared keys / the setup behind every example

The sender and receiver share a secret key before exchanging messages.

All five ciphers here use symmetric-key encryption: the decryption operation is determined by the same shared key.

An interceptor who learns that key can also decrypt. Getting the key to the receiver securely is a separate problem.

Encryption aims to protect confidentiality; it does not by itself guarantee authenticity.

A minimal threat model

Attacker’s information	What can be attempted?
Ciphertext only	Infer patterns using a language model.
Known plaintext + matching ciphertext	Recover key constraints directly.
Chosen plaintext	Observe encryption of selected messages.

We mainly use ciphertext-only attacks today; known plaintext gives useful contrasts.

Always say what the attacker can see or request.

Our notation and conventions

- Use the 26-letter English alphabet: A = 0, B = 1, ..., Z = 25.
- Examples are normalized to uppercase; spaces and punctuation are removed.
- Indices start at 0 unless a table explicitly uses positions 1, 2, … .
- For each cipher, define any padding or filler rule before using it.

Normalization loses information before encryption; it is not encryption.

Arithmetic on a circular alphabet

WRAP AROUND

$Y + 2 \to A$ $\qquad$ $A - 2 \to Y$

Modulo 26 keeps the result in $\{0, 1, \ldots, 25\}$.

NUMERIC CHECK

$$(24 + 2) \bmod 26 = 0$$
$$(0 - 2) \bmod 26 = 24$$

In code, ensure negative remainders are normalized to 0–25.

SHIFT

01 / Shift cipher

One secret offset for every letter.

KEY

$K \in \{0, \ldots, 25\}$

We include K = 0 mathematically; it leaves the message unchanged.

The Caesar cipher conventionally uses K = 3.

Encrypt and decrypt a shift

ENCRYPT

$$C[i] = (P[i] + K) \bmod 26$$

DECRYPT

$$P[i] = (C[i] - K) \bmod 26$$

The same shared key K controls both operations.

Worked example / K = 2

PLAINTEXT

CRYPTOGRAPHYISFUN

CIPHERTEXT

ETARVQITCRJAKUHWP

Wrap-around checks: Y (24) → A (0); Z (25) → B (1).

Check both ordinary letters and wrap-around cases.

Attack the shift cipher

Try each key and inspect the candidate plaintext.

Candidate K	Decrypt KHOORZRUOG
1	JGNNQYQTNF
2	IFMMPXPSME
3	HELLOWORLD
4	GDKKNVNQKC

Only 26 keys exist, including the identity key. Exhaustive search is tiny.

The successful candidate is HELLOWORLD at K = 3.

Try it / a shift challenge

Pair exercise • 3 minutes

CIPHERTEXT

```
PHHWPHDIWHUFODVV
```

1. Recover the plaintext without being given K.
2. Explain how you know your answer is plausible.
3. If you know plaintext M corresponds to ciphertext P, what is K?

Work first; the next slide reveals the answer.

Debrief / little search, useful structure

SOLUTION

$K = 3 \rightarrow \text{MEETMEAFTERCLASS}$

KNOWN-PLAINTEXT CHECK

$K = (15 - 12) \bmod 26 = 3$

Readable language helps select a key. Very short strings may admit several plausible readings.

An attacker need not recover the key by brute force if a matching letter is known.

02 / Monoalphabetic substitution

Replace the fixed shift with an arbitrary alphabet permutation.

Each plaintext letter still always becomes the same ciphertext letter.

More keys—but the language still leaves fingerprints.

The key is a bijection

PLAINTEXT ALPHABET

```
ABCDEFGHIJKLMNOPQRSTUVWXYZ
```

SUBSTITUTION ALPHABET

```
QWERTYUIOPASDFGHJKLZXCVBNM
```

WORKED EXAMPLE

```
BECAUSE → WTEQXLT
```

Decryption uses the inverse table. Distinct input letters must have distinct outputs.

A repeated output letter would make unique decryption impossible.

A huge key space is not enough

NUMBER OF TABLES

$$26! \approx 4.03 \times 10^{26} \approx 2^{88.4}$$

At one billion keys per second, trying every table would take about 12.8 billion years.

Yet a substitution preserves letter counts and repeated-letter patterns.

An attack can exploit structure rather than enumerate every key.

What survives substitution?

PATTERN

MEET → DTTZ

ANOTHER PATTERN

LETTER → STZZTK

The two middle letters remain equal.

Equalities survive; letter identities change.

Word boundaries, if retained, reveal even more. We remove them in our main example.

A bijective substitution relabels the frequency histogram.

BREAK 01

Break / return in 10 minutes

Before you go: why does 26! fail to guarantee security?

After the break, we will use the surviving patterns to attack a complete message.

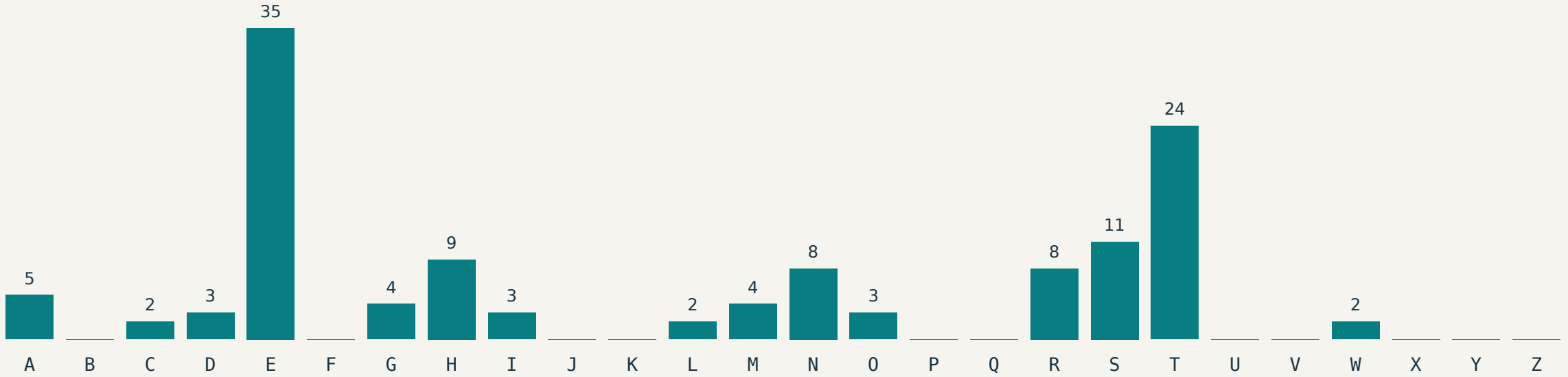
Elapsed time: 50–60 minutes

Frequency analysis / evidence, not certainty

- Count letters in the ciphertext and compare with a plausible language model.
- Use pairs, triples, repeated patterns, and context to test candidate mappings.
- Revise guesses when they contradict other evidence.

The most frequent ciphertext letter is not guaranteed to represent E.

A histogram with the labels scrambled



Plaintext letter labels • bar height = count; substitution preserves every bar height

This is the actual distribution of our example—not a universal English frequency table.

The substitution changes the labels, while the counts stay exactly the same.

The intercepted message

123 LETTERS • SPACES REMOVED

```
ZITLTEKTZDTTZOFUOLOFZITTQLZZGVTKZITDTLLTFUTKFTTR
LZITLTEKTZSTZZTKLTFRZITSTZZTKZGZITTQLZZGVTKQFRZI
TFDTTZZITDTLLTFUTKQZZITUQZT
```

Assume a single substitution table was used throughout.

This deliberately repetitive example is not representative English prose.

Count first; guess second

Ciphertext symbol	Count	Relative frequency
T	35	28.5%
Z	24	19.5%
L	11	8.9%
I	9	7.3%
K	8	6.5%
F	8	6.5%

T is a reasonable E candidate. Other mappings need additional evidence.

Message topic, length, and style can shift the expected ranking.

Use repeated triples to test a guess

REPEATED CIPHERTEXT PATTERN

ZIT

Hypothesis: ZIT represents THE. Then Z → T, I → H, and T → E.

A CONSEQUENCE TO CHECK

ZITLTEKTZ → THESECRET

A plausible completion is THESECRET. This adds L → S, E → C, and K → R.

A word-shaped guess must agree with every occurrence of its letters.

Reveal only the letters supported so far

MAPPINGS USED

Z→T I→H T→E L→S E→C K→R

PARTIAL DECRYPTION

```
THESECRET.EET....S..THEE.STT..ERTHE.ESSE..ER.EE.
STHESECRET.ETTERSE..THE.ETTERT.THEE.STT..ER...TH
E..EETTHE.ESSE..ER.TTHE..TE
```

Repeated fragments suggest further word hypotheses.

Try a new mapping everywhere before accepting it.

Here we show plaintext letters in uppercase and unknown letters as dots.

Exercise / grow the partial key

Small-group exercise • 4 minutes

LOOK FOR THESE FRAGMENTS

DTTZ0FU DTLLTFUTK STZZTK

Use the current partial key to decode what you can.

Propose two additional letter mappings and justify them with context.

Which of your conclusions are evidence-based guesses rather than certainties?

The next slide shows the full teaching plaintext.

Debrief / reconstruct the message

THE SECRET MEETING IS IN THE EAST TOWER. THE MESSENGER NEEDS THE SECRET LETTER. SEND THE LETTER TO THE EAST TOWER AND THEN MEET THE MESSENGER AT THE GATE.

EXAMPLE NEW MAPPINGS

D→M O→I U→G S→L F→N

The recovered letters must also produce a consistent one-to-one substitution.

Letters absent from the message may remain undetermined even after reading it.

When simple frequency analysis struggles

- Short messages: counts are noisy, and many readings can fit.
- Unusual vocabulary or a different language: the assumed model may be wrong.
- Missing spaces: word boundaries must be inferred.
- Longer language patterns help: pairs, triples, and four-letter sequences.

More ciphertext often helps—but there is no universal length that guarantees success.

03 / Vigenère cipher

Use several shifts, selected by a repeating keyword.

KEY IDEA

`Position in the key changes the mapping.`

The weakness moves from one fixed mapping to a repeating schedule.

A repeating key selects the shifts

RULE

$$C[i] = (P[i] + K[i \bmod m]) \bmod 26$$

WORKED EXAMPLE

```
Plaintext   ATTACKATDAWN
Key         LEMONLEMONLE
Ciphertext  LXFOPVEFRNHR
```

The key length is m = 5. Decryption subtracts the corresponding key value.

The two A letters at positions 0 and 3 encrypt differently.

Exercise / encrypt, then reverse

Pair exercise • 3 minutes

PLAINTEXT

CRYPTOGRAPHY

KEY

LUCK

Repeat the key, encrypt the first six letters, and check the wrap-around.

Then decrypt your result by subtracting the same repeated key.

A = 0; the first key shift L is 11, not 12.

Debrief / check every column

ALIGNED ROWS

```
Plaintext   CRYPTOGRAPHY
Key         LUCKLUCKLUCK
Ciphertext  NLAZEIIBLJJI
```

FIRST TWO LETTERS

```
C + L: (2 + 11) mod 26 = 13 → N
R + U: (17 + 20) mod 26 = 11 → L
```

One plaintext letter can have different ciphertext letters.

The attack has two stages

Stage	Question	Tool
1. Find a plausible period	When does the shift schedule repeat?	Kasiski; index of coincidence
2. Recover each shift	Which Caesar shift fits each column?	Language scoring; trial decryption

The effective period can be shorter than the written keyword: ABCABC repeats every three letters.

Finding a candidate key length is a hypothesis, not the end of the attack.

Kasiski / repetitions reveal distances

PLAINTEXT

THESUNANDTHEMANINTHEMOON

KEY

KINGKINGKINGKINGKINGKING

CIPHERTEXT

DPRYEVNTNBUKWIAOXBUKWWBT

The plaintext THE at positions 9 and 17 has the same key alignment: distance 8.

Both occurrences encrypt to BUK. The period 4 divides their distance.

Positions are zero-based. Repeated segments can also occur by coincidence.

Kasiski / turn distances into candidates

Find repeated ciphertext segments, usually at least three letters long.

Measure distances between their starting positions and look for common factors.

ILLUSTRATIVE DISTANCE SET

```
Distances: 30, 45, 75
gcd(30, 45, 75) = 15
Candidate periods: 1, 3, 5, 15
```

Coincidences can spoil a gcd; true repeats may reveal a multiple of the period.

Use several repeats and cross-check the candidate periods statistically.

Index of coincidence / count collisions

Pick two distinct positions at random. What is the chance their letters match?

DEFINITION

$$IC = \Sigma\ f[a](f[a] - 1) / (N(N - 1))$$

f[a] is the count of letter a; N is the number of letters. Require N ≥ 2.

SMALL CHECK

AAB: IC = 2 / 6 = 1/3

The sum is over all 26 letters; ordered pairs are counted consistently.

What IC does—and does not—measure

Text model	Expected / typical IC
Independent uniform letters	1/26 ≈ 0.0385
Ordinary English prose	Often about 0.065; varies by text
One fixed substitution of that prose	Exactly the same IC as the original text

IC is a statistical clue, not a general test of security.

A Caesar shift preserves counts up to relabeling, so it preserves IC.

Guess a period; split into columns

FOR CANDIDATE PERIOD m = 5

```
Column 0: C[0], C[5], C[10], …
Column 1: C[1], C[6], C[11], …
…
Column 4: C[4], C[9], C[14], …
```

At the correct period, each column is encrypted by a single fixed shift.

Wrong periods usually mix shifts. Multiples of the true period may also look good.

Short columns make IC estimates unreliable; prefer enough data and corroborating evidence.

A reproducible IC experiment

Candidate period	Mean column IC	Interpretation
1	0.0429	Mixed key positions
2	0.0429	Mixed key positions
3	0.0428	Mixed key positions
4	0.0432	Mixed key positions
5	0.0668	Strong candidate / multiple
10	0.0684	Strong candidate / multiple

Computed from 1045 letters of original English prose, encrypted with key LEMON.

The full plaintext, ciphertext, and calculations accompany this deck.

Recover a column by scoring all 26 shifts

For each candidate shift, decrypt the column and compare its letter counts with an English model.

ONE STANDARD SCORE

$$\chi^2 = \Sigma\ (\text{observed} - \text{expected})^2\ /\ \text{expected}$$

Lower scores indicate a closer fit to that model; try promising candidates in the complete message.

Use every letter count—not only the most frequent letter.

Enumerating 26 candidates per column is practical; language scoring makes it useful.

Recover the key / confirm the message

Column	Best-scoring shift	Key letter
0	11	L
1	4	E
2	12	M
3	14	O
4	13	N

Recovered key: LEMON. Decryption begins THEHISTORYOFSECRETWRITING…

We do not need to enumerate all 26^m complete keys: columns can be scored independently.

Checkpoint / what broke Vigenère?

Discussion • 3 minutes

Explain why repeating the key creates several interleaved shift ciphers.

Why can a candidate period of 10 look convincing when the true period is 5?

What extra uncertainty would you expect from a 25-letter message?

Answers: periodic alignment; multiples preserve alignment; too little data per column.

BREAK 02

Break / return in 10 minutes

So far: a huge table leaks frequencies; a repeating key leaks periodicity.

Next: encrypt pairs of letters, then try moving the letters instead.

Elapsed time: 110–120 minutes

04 / Playfair cipher

Encrypt pairs of letters using a small key square.

Invented by Charles Wheatstone in 1854 and promoted by Lyon Playfair.

A practical historical hand cipher—not a present-day security mechanism.

From codebooks to letter pairs

A codebook can assign a code group to an entire word or instruction.

CODEBOOK ILLUSTRATION

“Attack at dawn” → 48271

Playfair instead uses a compact rule to substitute letter pairs. It avoids distributing a huge phrase table.

Historical attraction: manual operation with more complexity than a single-letter substitution.

Codebooks operate on meaningful units; Playfair operates on prepared letter pairs.

Build the Playfair square

Keyword: PLAYFAIR EXAMPLE

1. Merge J into I.

2. Remove spaces and repeated letters, keeping first occurrences.

3. Append unused letters in alphabetical order.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Distinct keyword letters: P L A Y F I R E X M. The square has 25 cells.

Prepare pairs carefully

RULE

Scan left to right; form pairs of distinct letters.

If the next two letters match, insert X after the first and reconsider the second letter.

Pad a final single letter with X. For a repeated or final X, use Q instead.

EXAMPLE

BALLOON → BA LX LO ON

X → XQ XX → XQ XQ

This lecture specifies Q as the alternate filler; implementations can use different conventions.

Three geometric rules

Same row: move each letter one cell right, wrapping at the edge.

Same column: move each letter one cell down, wrapping at the bottom.

Rectangle: keep each letter’s row and use the other letter’s column.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Apply exactly one rule to each prepared pair.

Worked pairs / see each rule

SAME ROW

PL → LA

SAME COLUMN

PI → IB

RECTANGLE

HI → BM

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

For a rectangle, preserve the order of the two input rows.

A complete Playfair example

NORMALIZED PLAINTEXT

HIDETHEGOLDINTHETREESTUMP

PREPARED PAIRS

HI DE TH EG OL DI NT HE TR EX ES TU MP

ENCRYPTED PAIRS

BM OD ZB XD NA BE KU DM UI XM MO UV IF

Notice the inserted X in TREE: preparation changes the representation before encryption.

Standard PLAYFAIR EXAMPLE illustration, also used in the supplied deck.

Decrypt by reversing the movement

Same row: move left.

Same column: move up.

Rectangle: use the same opposite-corner rule.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

CHECK

BM → HI
LA → PL
IB → PI

Decryption recovers the prepared text, including any inserted fillers.

Exercise / decrypt a Playfair message

Pair exercise • 4 minutes

KEYWORD

PLAYFAIR EXAMPLE

CIPHERTEXT

RO NC FL WI

Use the same square and write down the rule used for each pair.

After decryption, decide whether any filler should be removed.

The answer follows on the next slide.

Debrief / fillers require interpretation

PAIR-BY-PAIR SOLUTION

```
RO → EN     NC → CR
FL → YP     WI → TX
```

PREPARED PLAINTEXT

ENCRYPTX

The prepared plaintext is ENCRYPTX. The final X pads the last single letter.

The intended reading is ENCRYPT. We remove the final X using context, not a cipher rule.

Keep the padding in the mechanical decryption; interpret it in a separate step.

PLAYFAIR

One more challenge / repeated letters

Pair exercise • 2 minutes

CIPHERTEXT

DP YR AN QO

Decrypt with the same square. Then infer a likely original word.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Solution: BA LX LO ON → BALLOON. The inserted X splits the original repeated L.

Why filler removal is ambiguous

EXAMPLE

```
AXA → AX AX
AA  → AX AX
```

Both messages prepare to AXAX under our rules, so they encrypt identically.

Removing every X would corrupt genuine X letters. Merging I/J also loses information.

The cipher is reversible on prepared pairs; preparation is not reversible in general.

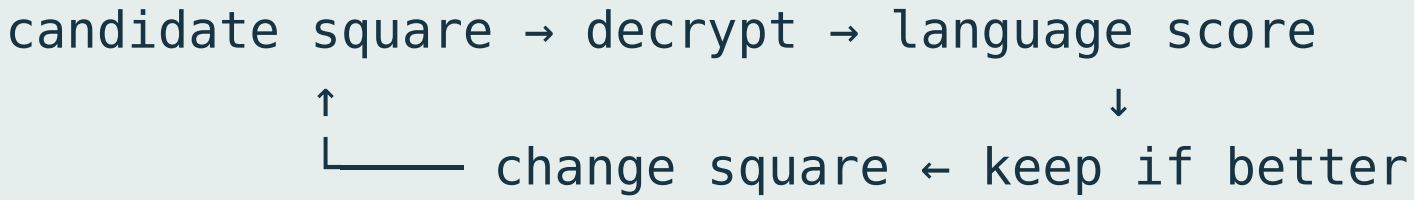
How Playfair can still be attacked

- Single-letter frequencies are less direct, but pair and longer patterns survive.
- Try a candidate square; decrypt and score the result using language statistics.
- Change the square and retain better candidates. Use restarts to escape poor local choices.

Search quality depends on text length, language, scoring, and computation—not a fixed guarantee.

Hill climbing / the attack as an algorithm

SEARCH LOOP



For example, swap two square entries and compare four-letter sequence scores.

A local optimum may not be the correct square. Multiple random starts help.

A readable result must be checked by re-encryption and contextual consistency.

05 / Transposition cipher

Keep the letters. Change only their positions.

FOCUS

`Fixed-size block permutation`

A permutation is a one-to-one rearrangement of positions.

Define the direction of the permutation

Ciphertext position i	1	2	3	4	5	6
Plaintext position π(i)	3	1	4	6	2	5

RULE

$$C[i] = P[\pi(i)]$$

This table tells us which plaintext position supplies each output position.

The block size is 6. Pad the last block with X; retain the original length separately.

Some sources define the reverse direction; a worked example prevents ambiguity.

Worked example / rearrange two blocks

PLAINTEXT

```
ATTACK ATDAWN
```

PERMUTATION

```
Read positions 3, 1, 4, 6, 2, 5
```

CIPHERTEXT

```
TAAKTC DAANTW
```

First block: input positions 3, 1, 4, 6, 2, 5 give T, A, A, K, T, C.

No letter changes identity; every block retains exactly its original letter counts.

Decryption uses the inverse permutation

ENCRYPTION ORDER

$\pi = [3, 1, 4, 6, 2, 5]$

DECRYPTION ORDER

$\pi^{-1} = [2, 5, 1, 3, 6, 4]$

CHECK

TAAKTC → ATTACK

Equivalently, place each ciphertext letter back into the plaintext position named by the encryption table.

Verify the direction by checking one full round trip.

Exercise / the letters are all still there

Pair exercise • 3 minutes

PLAINTEXT

DEFEND

Encrypt with π = [3, 1, 4, 6, 2, 5]. Then decrypt your result.

How many possible permutations are there for this block size?

Would ordinary single-letter frequency counts change?

Check repeated letters as well as the table direction.

Debrief / preserved counts expose structure

SOLUTION

DEFEND → FDEDEN

There are 6! = 720 possible keys for this particular block size.

The histogram is unchanged. Candidate rearrangements can be ranked by language patterns.

With longer blocks, exhaustive search grows, but preserved statistics remain a weakness.

Permutation is a useful design ingredient; this simple cipher alone is weak.

SYNTHESIS

Compare the five ciphers

Cipher	Key / structure	Useful attack clue
Shift	One offset	26 candidates; readable text
Substitution	One alphabet permutation	Counts and repeated patterns
Vigenère	Periodic shifts	Repetitions; column statistics
Playfair	5 × 5 square; letter pairs	Language-scored square search
Transposition	A fixed position permutation	Unchanged counts; word patterns

For each design, ask: what does the transformation fail to hide?

SYNTHESIS

A contemporary reason to care

A team stores an API credential in a configuration file, shifted by 13 letters, and calls it “encrypted”.

An attacker who reads the file can try every shift immediately. Hiding the transformation offers little help.

The engineering question is who can obtain the secret—and what assumptions protect it.

Illustrative present-day scenario, not a reported incident. Never protect real secrets with these ciphers.

SYNTHESIS

From letters to exact digital data

BYTES • HEXADECIMAL

```
A → 41      0 → 30      ? → 3F      space → 20
```

These are ASCII code values represented as bytes; ASCII itself is a 7-bit code.

Modern encryption accepts bytes or bits, so spaces, files, and images can be recovered exactly.

Security needs more than obscuring letter frequencies: key handling and protection against tampering also matter.

We motivate later lectures here without introducing additional cipher algorithms.

SYNTHESIS

Exit questions / explain your reasoning

Individual reflection • 4 minutes

1. Why can a cipher with 26! keys still be breakable?
2. Which observation makes a repeating-key cipher vulnerable?
3. Why can Playfair decryption fail to recover the exact original spelling?
4. Which cipher leaves the complete letter histogram unchanged?

A useful answer identifies a surviving property, not just an attack name.

SYNTHESIS

Exit discussion / the four ideas

- Substitution preserves frequency structure even when exhaustive search is infeasible.
- Vigenère repeats the shift schedule; aligned columns reduce to shift ciphers.
- Playfair preparation inserts fillers and merges I/J, losing information.
- Transposition leaves letter identities—and therefore their counts—unchanged.

All five ciphers are learning tools, not recommendations for real-world encryption.

REFERENCES

Reading and sources

Primary teaching reference: supplied *Lecture02_Classical_Ciphers.pdf*, Wu Hongjun, 79 pages. Scope and example-based approach retained; examples and explanations revised.

Technical reading: Stinson & Paterson, *Cryptography: Theory and Practice*, 4th ed., Chapter 1. Section numbering differs across editions.

History: Kahn, *The Codebreakers* (rev. ed., 1996); Kerckhoffs (1883); Kasiski (1863).

Companion materials: teaching guide, reproducible examples, and build source accompany this PDF. No external images or copied news articles are used.

The lecture ends here. All numerical examples in this deck are generated and checked in the build source.